A Drop is All It Takes: Data Reconstruction from Partial Transformer Activations in Distributed Machine Learning

Anonymous ACL submission

Abstract

Large Language Models and transformer-based architectures are increasingly deployed in privacy-sensitive distributed settings such as split learning, collaborative inference, and federated training, where intermediate activations are routinely exposed across trust boundaries. While prior privacy attacks have primarily focused on gradient inversion and forward-inversion methods, these findings suggest that intermediate hidden representations can also leak substantial information about the original input. This motivates our focus on analyzing activations in split learning and collaborative inference. Transformer activations can preserve lexical, syntactic, semantic, and positional information throughout the forward pass, making them a natural attack surface for reconstruction. In this work, we present a comprehensive layer-wise and sub-module-level analysis of activation leakage in transformer architectures, investigating whether individual linear components are independently sufficient for input reconstruction, demonstrating distinct leakage profiles for layers and various linear modules, highlighting how lexical and semantic meaning evolves during forward propagation. Our experiments indicate that significant training data leakage can occur from activations associated with a single Transformer layer, or even a single linear component containing only 0.92% of the model's parameters. We also demonstrate that applying differential privacy to activations during training provides only limited protection against data leakage. Our code will be made publicly available soon.

1 Introduction

Large Language Models (LLMs) have become foundational components in modern artificial intelligent systems, enabling significant advances in natural language understanding, generation, reasoning, and classification purposes. LLMs are increasingly deployed in privacy-sensitive domains including healthcare, finance, and enterprise analytics. Recent surveys have highlighted that despite their remarkable capabilities, LLMs introduce substantial security and privacy risks arising from memorization, information leakage, adversarial prompting, and distributed training vulnerabilities (Das et al., 2025).

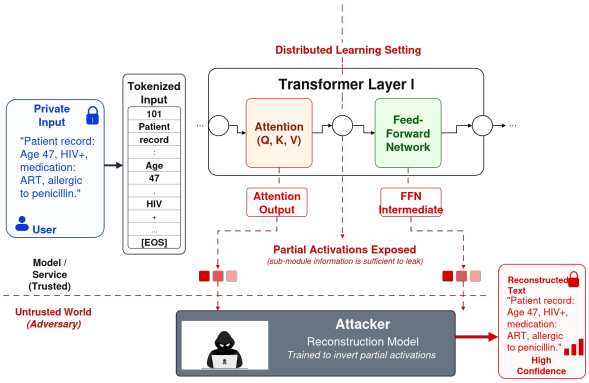


Figure 1: Overview of Data Reconstruction Attacks via Partial Activations. An attacker intercepts intermediate activations across different sub-modules during the forward pass of a Transformer layer in a split setting to reconstruct the private user input.

The models are advancing substantially in the number of parameters as well as dataset, to address the computational and ownership challenges associated with deploying large-scale models, Federated learning (FL) and split learning (SL) have emerged as the dominant framework for distributed training with privacy guarantees, operating under the premise that sharing intermediate representations rather than raw data provides meaningful protection (McMahan et al., 2017; Gupta and Raskar, 2018). In SL, models are partitioned across multiple parties, and only intermediate activations commonly referred to as smashed data, are transmitted instead of raw user inputs. This design is often assumed to provide a degree of privacy because the original training samples remain local to the client.

This premise has been progressively eroded,

with the full model gradients being sufficient to reconstruct private training inputs with high fidelity (Zhu et al. 2019), a result since extended by various attacks using forward activations and backward gradients in distributed settings. Most recently, it has been onserved that partial gradients are sufficient to reconstruct private text, effectively eliminating the defense value of parameter freezing and layer-wise training (Li et al., 2024).

Nevertheless, many modern distributed inference and split-learning deployments expose intermediate hidden representations. Recent work on split learning for LLM fine-tuning demonstrated that smashed activations themselves may provide strong priors for reconstruction attacks, particularly due to the "not-too-far" property of fine-tuning and the highly structured representations learned by transformer-based language model. Along with the growing class of deployment settings such as Inference-only APIs, split-inference pipelines, and model-as-a-service frameworks (Chen et al., 2024) expose forward intermediate activations to the server. The existing studies largely evaluate leakage at coarse granularity, focusing on complete hidden states or combined layer outputs, leaving an important question insufficiently explored.

RQ. *To what extent do individual transformer sub-modules independently preserve reconstructible information about the original input?*

Our study is motivated by the observation that transformer representations are highly structured throughout the forward propagation channel. Even intermediate linear transformations preserve semantic, syntactic, and positional characteristics of the original text. By isolating activations from individual sub-modules and training dedicated inversion models for each capture point, we demonstrate that significant input reconstruction is possible even from isolated linear projections without access to gradients, parameters, or model outputs. Our key contributions include:

- **Comprehensive Sub-Module Vulnerability Mapping:** We formalize a systematic, layer-wise evaluation framework to quantify information leakage across distinct internal linear components of transformer architectures, including query, key, value, and attention output projections, as well as feed-forward and embedding layers. By successfully extending the Semi-white-box Forward Inversion Paradigm (SIP) (Chen et al., 2024), we map these structural data liabilities comprehensively across the entire BERT transformer family.

- **Characterization of Asymmetric Leakage Profiles:** We introduce a fine-grained vulnerability mapping of transformer inference that localizes reconstruction risk to specific internal components. This analysis identifies early-layer Query/Key projections and FFN expansion activations as high-risk leakage points, while distinguishing them from comparatively less vulnerable integrated attention-output representations.

- **Empirical Evaluation of Perturbed Defenses:** We design and execute a benchmark evaluating the efficacy and operational limits of activation-level DP-Forward defense against forward layer-wise data inversion attacks.

2 Background and Related Work

2.1 Split Learning and Intermediate Activation Exposure

Split Learning (Gupta and Raskar, 2018; Vepakomma et al., 2018) enables distributed training by partitioning a model across a client and a server, with only intermediate activation vectors, the "smashed data" transmitted at the cut layer. The U-shaped variant extends this to a three-segment (head–trunk–tail) architecture where the trusted client holds both the head and tail, and the server manages the parameter-intensive trunk (Lyu et al., 2023). During forward computation, the client processes raw data, generates smashed data, and sends it to the server for trunk layer training. The client then trains the tail layers to infer labels. Given an input token sequence $x = (x_1, \ldots, x_S)$, the client computes

$$\tilde{x} = f_{\text{head}}(x) \in \mathbb{R}^{B \times S \times H},$$

and transmits only $\tilde{x}$ to the server. The raw input x remains local to the client. where B represents the batch size, S is the sequence length, and H is the hidden dimension, and transmits $\tilde{\text{x}}$ to the server. The key privacy claim of SL is that x is never exposed directly only the intermediate $\tilde{\text{x}}$ leaves the client. Recent line of work has systematically countered this claim, demonstrating that $\tilde{\text{x}}$ is a far richer signal than originally assumed (Chen et al., 2024; Liu et al., 2025).

2.2 Reconstruction Attacks and Privacy Leakage in Language Models

The broader literature on data reconstruction for various transformer based language models (Zhu et al., 2019; Balunovic et al., 2022) has focussed primarily on the federated-learning setup where the honest but curious server observes the complete gradients $\nabla_\theta \mathcal{L}(x, y)$ and attempts to recover (x, y) by minimizing the gradient comparison loss.

DLG (Zhu et al., 2019) first demonstrated this for image models; TAG (Deng et al., 2021) extended the approach to transformer language models using a combined $L1 + L2$ distance metric; and LAMP (Balunovic et al., 2022) advanced reconstruction through selective initialization, embedding regularization, and token reordering guided by a prior language model.

The Mix and Match (Yang, 2026) approach investigated reconstruction in a LLM-based text classifier regime. Their attack couples the backbone's generative prior with the classifier head's discriminative guidance to produce a joint objective:

$$J(\mathbf{x}) = \alpha \cdot \log P_{\text{base}}(\mathbf{x}) + \beta \cdot g(\mathbf{x}) - \gamma \cdot R(\mathbf{x}),$$

where

$$g(\mathbf{x}) = \log f_{y^*}(\mathbf{x})$$

is the classifier guidance score for target label y^*, and $R(\mathbf{x})$ is a regularization term. Demonstrating that reconstruction risk extends to classification architectures that expose no generative decoding endpoint, guiding that forward activations from individual projection layers are sufficient for reconstruction without any generative output.

2.3 Forward Activation Inversion Attack

The vulnerability of forward intermediate representations to inversion was first characterized for collaborative inference on image models (Zhang et al., 2023). For language models, the semi-white-box Forward Inversion Paradigm (SIP) (Chen et al., 2024) establishes that a learned inversion model specifically a bidirectional GRU decoder, trained on an auxiliary dataset can map the smashed data $\tilde{\mathrm{x}}$ directly to token predictions.

Formally, SIP trains an inversion decoder d_ϕ on a public auxiliary dataset D_a. For an auxiliary input sequence $b \sim D_a$, the attacker first computes a surrogate smashed representation

$$\tilde{b} = \bar{f}_{\text{head}}(b),$$

where $\bar{f}_{\text{head}}$ is a frozen surrogate of the client-side head model. The decoder is then optimized using token-level cross-entropy:

$$\mathcal{L}_{\text{AE}}(\phi) = \mathbb{E}_{b \sim D_a} \left[\text{CE} \left(d_\phi(\bar{f}_{\text{head}}(b)), b \right) \right].$$

Here, $d_\phi(\bar{f}_{\text{head}}(b))$ produces token logits for each position in the sequence, L_{AE} being the auto-encoder reconstruction loss, and $\text{CE}(\cdot, \cdot)$ denotes the cross-entropy between the predicted token distribution and the original auxiliary sequence b.

During this training stage, the parameters of $\bar{f}_{\text{head}}$ are fixed, and only the decoder parameters ϕ are optimized. This forces d_ϕ to learn a mapping from intermediate representations back to the original token sequence. At attack time, the adversary observes the real smashed representation

$$\tilde{x} = f_{\text{head}}(x),$$

where x is the private input sequence. The trained decoder then produces token-level logits,

$$\hat{Y} = d_\phi(\tilde{x}),$$

and the reconstructed text is obtained by token-wise decoding:

$$\hat{x}_s = \arg\max_{v \in \mathcal{V}} \hat{Y}_{s,v}.$$

In the original SIP setting, the decoder is trained for a single split point, namely the boundary between the client-side head and the server-side trunk. Our work extends this idea to internal transformer sub-modules. Instead of only considering the smashed representation $\tilde{x}$, we later consider activations $a_{m,l}(x)$ captured from a specific sub-module m at layer l, such as the query, key, value, attention-output, or feed-forward projection.

SIP in its original formulation evaluates leakage only at a single externally chosen split point. Consequently, it does not examine whether activations at intermediate sub-module boundaries within a transformer block independently carry sufficient signal for reconstruction. This sub-module-level leakage is the primary gap addressed in our work.

2.4 Defenses Against Activation Level Leakage

To prevent attackers from reconstructing original data from forward smashed data, mainly two types of defense strategies are explored:

- **Perturbation-based** approaches are commonly used (Mai et al., 2023), these methods obscure

sensitive information in private data with differential privacy (DP) (Dwork et al., 2006) noise during forward propagation.

- **Encryption-based** approaches offer rigorous security guarantees via mechanisms such as Homomorphic Encryption (Zhang et al., 2020) and Multi-Party Computation (Mugunthan et al., 2019), but suffer from high computational costs.

In this work, we also study the effectiveness of differential privacy (DP) as a defense against attacks on partially exposed activations. In forward propagation, noise is added to intermediate layers, such as embedding vectors (Chatzikokolakis et al., 2013) or activation values (Du et al., 2023), to prevent attackers from inferring original inputs from perturbed smashed data (Liu et al., 2025).

3 Methodology

This section presents the threat model and the attack structure used to take leverage of the intermediate activations for input construction

3.1 Threat Model

We consider an honest-but-curious setting commonly used in the split learning literature (Chen et al., 2024) where the server honestly follows distributed rules or training protocols but attempts to reconstruct private user inputs from exposed intermediate representations. The model is partitioned into a client resident head segment f_{head} spanning layers 0 through $k-1$, a server resident trunk f_{trunk} and optionally a client resident tail f_{tail}. During forward computation, the client processes its private input sequence $\mathbf{x} = (x_1, x_2, \ldots, x_S)$ and transmits the smashed data value $\tilde{\mathbf{x}} = f_{\text{head}}(\mathbf{x}) \in \mathbb{R}^{B\times S\times H}$ to the server as discussed previously. The **adversary capabilities** include:

1. White box access to the server side trunk model including its architecture and parameters.
2. Semi-white-box access to the client side head segment, which is the pre-trained weights f_{head} and the model architecture, the exact fine tuned weights of the client are not available.
3. Access to a public auxiliary dataset D_a WikiText in our experiments (Merity et al., 2016), whose distribution may differ from the private data of the client.
4. Intermediate activation values $a_{m,l}$ captured in a specific sub-module m in the transformer layer l during forward pass.

The adversary observes only the forward activation values; no gradient information, loss value, or label is available. The adversary does not perform any iterative optimization at inference time, the reconstruction is a single feedforward pass through a pretrained inverter.

Adversary Objective. For sub-module m in transformer layer l, let

$$a_{m,l}(x) \in \mathbb{R}^{B\times S\times H_m}$$

denote the activation exposed during the forward pass, where H_m is the output dimension of sub-module m. The adversary aims to recover

$$\hat{x} \approx x$$

using a learned inverter $d_\phi^{m,l}$:

$$\hat{x} = \text{Decode}\left(d_\phi^{m,l}(a_{m,l}(x))\right).$$

3.2 Hook-Based Activation Capture

Rather than modifying the split model architecture to expose activations at arbitrary sub-module boundaries, which would require separate model instances per capture point. We equip the model with "Forward Hooks" allowing a single pass to simultaneously populate activation values for all capture points. For each target sub-module $F_{m,l}$, we register a forward hook

$$h_{m,l} : (F_{m,l}, u, v) \mapsto S[(m,l)] \leftarrow \text{Extract}(v),$$

where u and v denote the input and output of the sub-module, S is a shared activation dictionary, and $\text{Extract}(\cdot)$ returns the primary tensor when the module output is a tuple.

3.3 The GRUDRInverter

The inversion model d_ϕ is a GRUDRInverter a lightweight sequence-to-sequence network consisting of a bidirectional GRU followed by a linear projection head. Given an intercepted activation

$$a_{m,l} \in \mathbb{R}^{B\times S\times H_m},$$

the inverter computes

$$z = \text{BiGRU}_\phi(a_{m,l}) \in \mathbb{R}^{B\times S\times 2d},$$

followed by a token-level linear projection

$$\hat{Y} = zW_{\text{out}} + b_{\text{out}} \in \mathbb{R}^{B\times S\times V},$$

where d is the hidden size of each GRU direction, V is the vocabulary size, $W_{\text{out}} \in \mathbb{R}^{2d \times V}$, and $b_{\text{out}} \in \mathbb{R}^{V}$. The bidirectional GRU processes each position in both forward and backward directions simultaneously, making use of the complete contextual structure of the activation values at each layer. The reconstructed token at position s is obtained as

$$\hat{x}_s = \arg\max_{v \in \mathcal{V}} \hat{Y}_{s,v}.$$

The inverter maps an activation tensor of shape (B, S, H_m) to token logits of shape (B, S, V), where H_m is the hidden dimension of the intercepted sub-module and V is the vocabulary size. Since most transformer sub-module activations preserve the batch and sequence dimensions (B, S), the same BiGRU-based inverter can be applied across different capture points by only adapting to the input feature dimension H_m.

4 Experimental Framework

We evaluate sub-module information leakage within internal transformer representations across progressive layer depths, isolated architectural matrices, and variable capacity constraints.

4.1 Experimental Settings

We benchmark the efficacy of the sub-module inversion framework across three distinct text classification datasets: the Stanford Sentiment Treebank (**SST-2**) (Socher et al., 2013), the Corpus of Linguistic Acceptability (**CoLA**) (Warstadt et al., 2019), and the **Rotten Tomatoes** movie review dataset (Pang and Lee, 2005), maintaining consistency with established data reconstruction literature. During model execution in a distributed setting, the target language processes their respective classification objectives while an honest-but-curious adversary intercepts isolated hidden representations to reconstruct the original token sequences.

To train the Semi-white-box Forward Inversion Paradigm (SIP) framework, we employ the **WikiText** corpus (Merity et al., 2016) as an out-of-domain auxiliary text distribution. This configuration ensures that the adversarial reconstruction decoder learns a generalized statistical prior of natural language syntax and vocabulary rather than overfitting to domain-specific label correlations belonging to the model specific target tasks. For evaluation, we randomly sample 10 independent text sequences from the validation or test splits of each classification corpus, maintaining a strict model execution batch size of 1 during inference to simulate realistic single-user query interceptions over exposed trust boundaries. We use dual NVIDIA Tesla T4 GPUs(GPU T4 x2) for our experiments.

4.2 Model Architecture and Capacity

To evaluate whether sub-module vulnerabilities scale with parameter density and representation width, our inversion experiments target three primary architectural and configurations of the transformer encoder:

- **$\text{BERT}_{\text{BASE}}$:** The standard frozen pre-trained `bert-base-uncased` configuration (Devlin et al., 2019) containing 12 transformer layers, 12 attention heads, and a hidden state width of $H = 768$ (110M parameters) to analyze baseline parameter vulnerability curves.
- **BERT_{FT}:** The task-adapted downstream implementation of the 110M parameter base architecture. We evaluated our reconstruction attacks against established, publicly available fine-tuned checkpoints from the Hugging Face repository (e.g., `textattack/bert-base-uncased-rotten_tomatoes`,`yoshitomo-matsubara/bert-base-uncased-cola`, and `textattack/bert-base-uncased-SST-2`). These models were trained following standard practices for GLUE benchmark tasks, typically 3 to 10 epochs using a learning rate of 2×10^{-5}, providing a robust baseline for evaluating the vulnerability of standard, off-the-shelf deployed models rather than custom-trained architectures.
- **TinyBERT:** A highly compressed distilled student model (`TinyBERT_General_4L_312D`) (Jiao et al., 2020) configured with 4 layers, 12 attention heads, and a down-scaled representation dimension of $H = 312$ (14.5M parameters) to study variations across condensed representation channels.

Rather than interacting with these architectures as monolithic blocks or restricting our evaluation to a single static client-server boundary, we map the complete internal attack surface layer by layer. We sweep across all 12 layers of $\text{BERT}_{\text{BASE}}$ and BERT_{FT} (Layers 0–11) and all 4 layers of TinyBERT (Layers 0–3) to trace how information leakage changes over the model processing depth. The number of activations in the models and their submodules are shown in Table 6.

Target Backbone	Evaluation Dataset	Intercepted Submodule Component	Layer Index	ROUGE-L	METEOR	Token Acc.
$\text{BERT}_{\text{BASE}}$	SST-2	`attn_query`	Layer 1	**0.942**	**0.915**	**0.948**
$\text{BERT}_{\text{BASE}}$	SST-2	`attn_key`	Layer 0	0.918	0.892	0.925
$\text{BERT}_{\text{BASE}}$	SST-2	`ffn_fc`	Layer 2	0.876	0.778	0.898
$\text{BERT}_{\text{BASE}}$	SST-2	`attn_output`	Layer 8	**0.535**	**0.459**	**0.723**
BERT_{FT}	Rotten Tomatoes	`attn_output`	Layer 1	**0.998**	0.934	**0.998**
BERT_{FT}	CoLA	`ffn_fc`	Layer 1	0.967	0.900	0.967
BERT_{FT}	SST-2	`attn_output`	Layer 1	0.779	0.723	0.832
BERT_{FT}	CoLA	`ffn_fc`	Layer 3	0.985	**0.958**	0.990
BERT_{FT}	SST-2	`attn_query`	Layer 3	0.912	0.841	0.923
BERT_{FT}	SST-2	`attn_output`	Layer 3	**0.695**	**0.606**	**0.795**
TinyBERT	CoLA	`attn_query`	Layer 0	0.945	0.912	0.951
TinyBERT	CoLA	`attn_value`	Layer 1	**0.963**	**0.927**	**0.953**
TinyBERT	Rotten Tomatoes	`attn_query`	Layer 0	0.921	0.895	0.933
TinyBERT	SST-2	`attn_qkv`	Layer 0	0.918	0.880	0.927
TinyBERT	SST-2	`attn_output`	Layer 3	**0.485**	**0.249**	**0.613**

Table 1: Representative inversion attack results across model architectures and sub-modules. Values represent ROUGE-L F1, METEOR, and Token Accuracy scores ($B = 1$). Results illustrate the high vulnerability of internal projection layers compared to integrated output modules, highlighting performance variance between Layer 1 and Layer 3. **Bold values** identify the maximum and minimum metrics within each backbone, emphasizing extreme structural bottlenecks (severe metric degradation) alongside contrasting instances of near-total information exposure.

4.3 Quantifying Reconstruction Measure

We evaluate the quality of the reconstructed sequences generated by our sub-module inversion decoders using the following alignment metrics:

- **ROUGE-L F1:** Computes the F-score of the longest common subsequence overlap between the original and reconstructed strings to evaluate the structural preservation of syntax (Lin, 2004).
- **METEOR:** Measures semantic alignment by incorporating exact token matches, stemming rules, and synonymy to evaluate recovery quality even when literal word forms diverge (Banerjee and Lavie, 2005).
- **Token Accuracy (*Token Acc*):** Evaluates the exact position-sensitive token recovery rate to establish a strict baseline for perfect sequence extraction.
- **Inversion Loss:** Evaluates the convergence behavior of the cross-entropy objective minimized by the adversarial decoder during proxy optimization.

4.4 Adversarial Sub-Module Targeting

Our attack framework extends the tracking principles of the Semi-white-box Forward Inversion Paradigm (SIP) (Chen et al., 2024) by equipping the model with forward hooks that tap into separate linear component outputs within an individual transformer block. Formally, for a given layer index, we isolate and intercept the forward activation vectors originating from two primary substructures:

1. **Attention Projections:** We evaluate intermediate activations immediately following the separate Query (Q), Key (K), and Value (V) weights, the fused QKV linear translation block, and the final multi-head attention integration block (`attn_output`).
2. **Feed-Forward Network (FFN):** We intercept the activations of the intermediate hidden expansion layer (`ffn_fc`), which scales features to a higher dimension, and the down-projection layer (`ffn_output`) which returns features back to the standard hidden width.

The reconstruction architecture is constructed as a bidirectional Gated Recurrent Unit (BiGRU) decoder optimized over the public auxiliary dataset for 2 epochs using a batch size of 4. During inference, the intercepted intermediate representation tensors are passed directly into the trained decoder to predict individual input tokens without needing access to any final classification head layers or generative text endpoints.

4.5 Framework for Perturbation Defenses

To establish an empirical baseline for defense optimization, we integrate an activation-level forward

Differential Privacy (DP) framework. We employ DP-Forward (Du et al., 2023) , adjusting the privacy budget ε while maintaining a clipping bound $C = 1.0$. To assess the effect of DP-noise on model utility, we fine-tune **BERT$_{\text{BASE}}$** on the **SST-2** dataset for **two** epochs and evaluate the resulting utility changes. We set the privacy parameter δ to 10^{-5} and consider privacy budgets $\varepsilon \in \{8, 4, 2\}$. After fine-tuning the model under the DP-Forward mechanism at the selected encoder layer 1, we evaluate its robustness by conducting the SIP attack using the same experimental setup as in the baseline configuration.

4.6 Experimental Results and Structural Analysis

Our modular evaluation demonstrates that forward activation inversion inside split learning environments exhibits highly nuanced, non-linear trajectories determined by internal transformation boundaries within model blocks.

Vulnerability Mapping over Processing Depth. As illustrated by the empirical curves in Figure 2, intermediate feature representations intercepted at early processing stages yield nearly perfect text reconstruction parameters. Intercepting forward activations at Layer 0 and Layer 1 of the baseline BERT$_{\text{BASE}}$ model exposes an absolute data liability. Specifically, the internal hidden states of the `attn_query` component at Layer 1 reach a peak ROUGE-L F1 score of 0.942 (Table 3), a METEOR score of 0.915, and an exact position-sensitive token accuracy of 0.948. This indicates that early transformation states preserve literal lexical identity vectors directly mapped to the underlying token embedding distributions.

As feature representations propagate into the intermediate sections of the architecture (Layers 4–8), the reconstruction fidelity remains stubbornly elevated. This indicates that despite iterative contextual transformations, individual component tensors independently preserve sufficient sequential constraints for successful extraction.

Only at the deepest blocks (Layers 10–11) we observe a gradual degradation in attack efficacy. This decline corresponds to abstract processing shifts where hidden states compress localized textual patterns into broader feature distributions optimized for the final task classifier head. Nevertheless, even at the absolute final transformation layer of BERT$_{\text{BASE}}$, the attack yields a high ROUGE-L F1 score, confirming that substantial token sequence

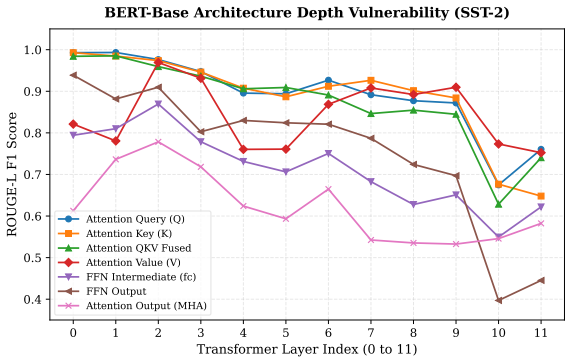


Figure 2: Granular tracking of forward reconstruction success (ROUGE-L F1) across all twelve layers and seven submodule linear projection boundaries of BERT$_{\text{BASE}}$ on the SST-2 dataset ($B = 1$).

Table 2: Evaluation of differential privacy defense on the **SST-2** dataset for varying privacy budgets on Encoder Layer 1 of **BERT$_{\text{BASE}}$**.

ε	ROUGE-L	METEOR	Model Acc.
8	91.61	92.60	82.34
4	89.96	90.90	82.00
2	96.54	97.19	**50.92**

data escapes across trust boundaries throughout the entire pipeline.

Asymmetric Sub-Module Vulnerability Profiles. A core contribution of our modular component tracking is identifying privacy vulnerabilities across internal linear matrices within the exact same transformer block. Rather than displaying uniform feature leakage, individual architectural transformations exhibit widely divergent leakage thresholds.

The primary information exposure vectors are concentrated within the raw linear projections (Table 3) specifically the Attention Query (Q), Attention Key (K), and the intermediate Feed-Forward expansion layer (`ffn_fc`). Conversely, the fully integrated multi-head attention output (`attn_output`) is consistently the most secure profile across the entire network depth.

This wide structural gap is caused by representation token fusion. Component vectors like Query, Key, and Value operate on isolated contextual variables before they undergo multi-head scaling and token dot-product mixing. The subsequent attention weighted summation acts as an architectural bottleneck that structurally fuses distinct tokens together, making sequence decryption significantly harder at the downstream `attn_output` stage. Sim-

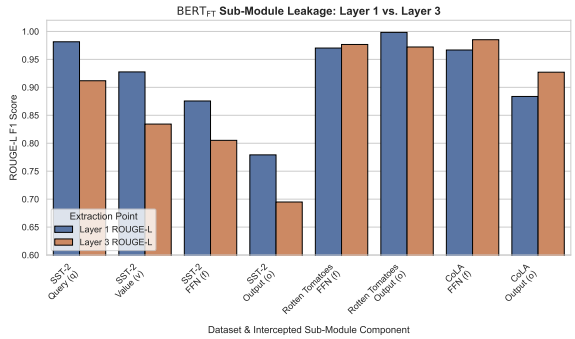


Figure 3: Reconstruction success (ROUGE-L F1) across downstream fine-tuned configurations (BERT$_{FT}$) comparing the degradation of micro-architectural components from Layer 1 to Layer 3 across evaluation datasets ($B = 1$).

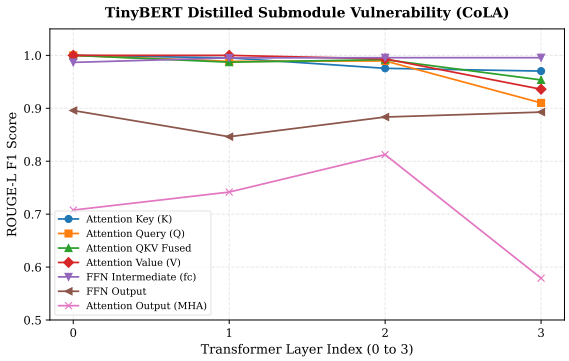


Figure 4: Layer-wise submodule reconstruction performance (ROUGE-L F1) for the compressed TinyBERT architecture across all four transformer layers on the CoLA dataset ($B = 1$).

ilarly, the intermediate expansion layer (`ffn_fc`) consistently leaks literal token structures at a higher rate than the final block output (`ffn_output`), acting as an internal warehouse of raw, uncompressed textual syntax.

Impact of applying DP-Forward. As shown in Table 2, enforcing stricter privacy budgets on encoder-layer 1 activations substantially reduces model utility. Furthermore, higher values of ε do not yield significant gains in attack mitigation, suggesting a limited privacy–utility trade-off in this setting.

Impact of Model Fine-Tuning. Crucially, downstream fine-tuning fails to restructure internal feature spaces, leaving early representation boundaries entirely exposed to adversarial reconstruction.

As shown in Table 1, Table 4 and Figure 3, intercepting superficial layers reveals an absolute vulnerability. On Rotten Tomatoes at Layer 1, the integrated `attn_output` yields a near-total plaintext recovery (ROUGE-L F1 of 0.998) (Table 5). Similarly, isolated key matrices (`attn_key`) on SST-2 at Layer 1 retain a severe leakage profile (0.991), proving early-stage embeddings remain undefended by downstream tuning.

Conversely, deeper layers (Layer 3) initiate metric degradation via representation fusion, though raw parameter boundaries stay uniquely exposed. On SST-2, attacking the isolated query vector (`attn_query`) yields an F1 score of 0.911 at Layer 3 (down from 0.981 at Layer 1), which drops further to 0.694 when targeting the integrated `attn_output` (down from 0.779 at Layer 1).

Meanwhile, intermediate expansion layers (`ffn_fc`) stay stubbornly high, shifting from 0.966 (Layer 1) to 0.985 (Layer 3) on CoLA (Table 4). This contrast confirms an asymmetric vulnerability: while depth and downstream tuning introduce noise to fused outputs like `attn_output`, the localized geometric axes within unfused linear projections remain heavily exposed.

Comparison Against Prior Inversion Baselines. While terminal black-box models bound ROUGE-L below 0.45 and the partial-gradient inversion of Li et al. (2024) caps token accuracy near 0.62, our modular tracking leads to much more reconstruction. Our results suggest that some sub-module activations allow highly accurate reconstruction. At Layer 1, `attn_output` on Rotten Tomatoes reaches a ROUGE-L and token accuracy of 0.998 (Table 5), outperforming the compared gradient-based baselines. Leakage also remains strong in deeper layers: at Layer 3, `ffn_fc` on CoLA achieves a ROUGE-L of 0.985, showing that feed-forward activations can still retain reconstructable token information.

5 Conclusion

This study demonstrates that transformer privacy in distributed environments is fragile, as micro-architectural analysis of intermediate activations enables high-fidelity sequence reconstruction. Experiments across BERT$_{BASE}$, BERT$_{FT}$, and TinyBERT demonstrate that attention and feed-forward layers persistently leak data, while the DP-Forward fine-tuning method fails to provide a privacy buffer. Consequently, these findings necessitate a shift in security focus from monolithic models toward topology-aware defenses designed to protect vulnerable linear projections.

Limitations

We try to provide an empirical study of activation-level leakage rather than a complete security analysis of all transformer deployments. Our experiments are limited to encoder-only BERT-family models, namely $\text{BERT}_{\text{BASE}}$, fine-tuned BERT, and TinyBERT, and to sentence-level classification datasets: SST-2, CoLA, and Rotten Tomatoes. We identify that leakage patterns may differ for decoder-only LLMs, encoder-decoder models, multilingual inputs, or domain-specific data such as medical, legal, or financial.

Our evaluation uses WikiText as public auxiliary data, a lightweight BiGRU inversion model, batch size one, and a small number of sampled evaluation sequences. These choices help isolate submodule leakage, but they do not fully capture real deployments with batching, padding, compression, repeated queries, or concurrent users. Stronger inversion models, larger auxiliary dataset, or different attacker setting may produce different reconstruction results.

We evaluate the defense effectiveness of DP- Forward and find that it is not adequate to mitigate the risk without significant model utility degradation. Therefore, further investigation into more effective defense mechanisms is necessary to improve the resilience of such systems against privacy attacks.

Ethics Statement

We use publicly accessible datasets and pretrained models for our experiments, strictly for academic purposes and in full accordance with their licensing terms. The findings suggest that intermediate activations should be treated as sensitive data, especially in high-risk domains such as healthcare, finance, education, legal services, and enterprise. Systems that share activations across trust boundaries should use appropriate safeguards, such as access control, cryptographic protocols, trusted hardware, or perturbation mechanisms.

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A Detailed Sub-Module Reconstruction Results

We report layer-wise and sub-module-level reconstruction fidelity under the Semi-white-box Forward Inversion Paradigm (SIP) across ROUGE-L, METEOR, Token Accuracy, and Reconstruction Loss. Entries in **bold** indicate near-perfect recovery (≥ 0.99 ROUGE-L).

Table 3: Reconstruction metrics on the **SST-2** dataset across $\text{BERT}_{\text{BASE}}$, TinyBERT, and BERT_{FT}.

Model	Target Component	Layer	ROUGE-L	METEOR	Token Acc	Loss
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	1	**0.9933**	0.9799	0.9936	12.1243
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	0	**0.9928**	0.9909	0.9947	10.9721
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	0	**0.9928**	0.9908	0.9947	10.8339
$\text{BERT}_{\text{BASE}}$	attn_qkv	1	0.9852	0.9758	0.9840	12.9471
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	1	0.9847	0.9875	0.9814	11.3401
$\text{BERT}_{\text{BASE}}$	attn_qkv	0	0.9841	0.9821	0.9836	12.0833
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	2	0.9763	0.9271	0.9740	12.2940
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	2	0.9738	0.9555	0.9808	17.2591
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	2	0.9693	0.9249	0.9740	11.1170
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_t$	0	0.9684	0.9467	0.9717	11.5204
$\text{BERT}_{\text{BASE}}$	word_emb	0	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	8	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	11	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	1	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	10	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	6	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	12	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	5	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	3	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	2	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	all_layers	4	0.9640	0.9242	0.9706	19.9294
$\text{BERT}_{\text{BASE}}$	attn_qkv	2	0.9593	0.9227	0.9654	11.7251
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	3	0.9474	0.8558	0.9396	12.2951
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	3	0.9462	0.8924	0.9514	13.5059
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	0	0.9387	0.8864	0.9485	14.6626
$\text{BERT}_{\text{BASE}}$	attn_qkv	3	0.9363	0.8360	0.9274	12.8432
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	3	0.9313	0.8312	0.9249	11.4450
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	6	0.9267	0.8472	0.9444	17.2867
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	7	0.9264	0.8665	0.9359	15.4636
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	6	0.9120	0.8156	0.9189	15.6287
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	9	0.9097	0.8101	0.9179	14.7935
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	2	0.9097	0.8420	0.9261	14.3549
$\text{BERT}_{\text{BASE}}$	attn_qkv	5	0.9091	0.8138	0.9216	15.0496
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	7	0.9080	0.8208	0.9176	15.2838
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	4	0.9075	0.8222	0.9127	14.6228
$\text{BERT}_{\text{BASE}}$	attn_qkv	4	0.9060	0.8240	0.9255	15.6319
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	8	0.9015	0.8513	0.9377	14.2055
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	4	0.8956	0.8255	0.9192	14.9647
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	5	0.8940	0.8165	0.9095	13.7028
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	8	0.8922	0.8325	0.9214	16.2591
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	7	0.8917	0.8163	0.9116	16.5438
$\text{BERT}_{\text{BASE}}$	attn_qkv	6	0.8910	0.8074	0.9150	15.6778
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	5	0.8867	0.8088	0.8974	14.3991
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	9	0.8840	0.8023	0.8984	14.6268
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	1	0.8815	0.7891	0.8912	20.1993
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	8	0.8773	0.7953	0.8994	15.4391
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	9	0.8719	0.8235	0.8944	14.0742
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	2	0.8694	0.7828	0.8809	19.8210
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	6	0.8686	0.7883	0.8950	15.0825
$\text{BERT}_{\text{BASE}}$	attn_qkv	8	0.8546	0.7483	0.8952	14.6653
$\text{BERT}_{\text{BASE}}$	attn_qkv	7	0.8464	0.7718	0.8968	14.8022
$\text{BERT}_{\text{BASE}}$	attn_qkv	9	0.8447	0.7377	0.8781	14.3790
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	4	0.8299	0.6863	0.8576	15.6335
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	5	0.8242	0.7053	0.8767	10.8229
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	0	0.8211	0.6904	0.8729	13.3899
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	6	0.8208	0.7115	0.8577	12.8337
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	1	0.8100	0.7184	0.8608	19.0400
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	3	0.8027	0.7006	0.8515	15.1811
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	0	0.7943	0.6839	0.8343	18.4078
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	7	0.7869	0.6694	0.8347	14.3944

Table 3 – continued from previous page

Model	Target Component	Layer	ROUGE-L	METEOR	Token Acc	Loss
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	1	0.7806	0.7097	0.8452	10.7776
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	3	0.7787	0.6329	0.8264	19.6493
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	2	0.7784	0.7111	0.8314	14.0436
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	10	0.7733	0.6786	0.8309	16.0913
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	5	0.7609	0.6942	0.8303	13.8016
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	11	0.7604	0.6598	0.8271	16.7975
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	4	0.7603	0.6716	0.8237	12.7569
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_v$	11	0.7523	0.6611	0.8288	16.4381
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	6	0.7505	0.6006	0.7879	20.5250
$\text{BERT}_{\text{BASE}}$	attn_qkv	11	0.7402	0.6420	0.8059	16.0761
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	1	0.7363	0.6508	0.8103	15.5021
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	4	0.7313	0.6081	0.8036	22.2883
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	8	0.7241	0.5924	0.8105	14.6474
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	3	0.7183	0.6186	0.8200	13.1529
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	5	0.7062	0.5923	0.7859	20.3370
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	9	0.6970	0.5131	0.7639	14.0273
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	7	0.6827	0.5481	0.8018	19.3676
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	10	0.6766	0.5953	0.7412	15.0050
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_q$	10	0.6748	0.5826	0.7633	15.8468
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	6	0.6647	0.6367	0.7945	13.6908
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	9	0.6509	0.5084	0.7672	18.9008
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_k$	11	0.6480	0.5478	0.7002	17.3182
$\text{BERT}_{\text{BASE}}$	attn_qkv	10	0.6284	0.5804	0.7215	14.8361
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	8	0.6277	0.5274	0.7510	20.2527
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	4	0.6240	0.5415	0.7792	14.3153
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	11	0.6220	0.4191	0.7180	19.8170
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	0	0.6126	0.5270	0.7050	14.2979
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	5	0.5934	0.4504	0.7201	13.0707
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	11	0.5823	0.4857	0.7193	14.5297
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_f$	10	0.5495	0.4204	0.7165	19.8065
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	10	0.5457	0.4319	0.6778	15.6166
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	7	0.5424	0.4389	0.6710	14.6116
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	8	0.5353	0.4592	0.7234	14.7790
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_o$	9	0.5326	0.4283	0.7107	14.2701
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	11	0.4452	0.3207	0.5513	14.8955
$\text{BERT}_{\text{BASE}}$	$\mathbf{A}_p$	10	0.3970	0.2932	0.5157	15.5943
TinyBERT	attn_qkv	0	**0.9948**	0.9802	0.9973	11.6018
TinyBERT	$\mathbf{A}_q$	0	**0.9933**	0.9801	0.9973	12.2821
TinyBERT	$\mathbf{A}_k$	0	**0.9912**	0.9827	0.9947	14.2311
TinyBERT	$\mathbf{A}_v$	2	0.9847	0.9769	0.9844	13.3280
TinyBERT	$\mathbf{A}_v$	0	0.9786	0.9742	0.9830	11.0108
TinyBERT	$\mathbf{A}_f$	0	0.9625	0.9560	0.9767	15.7584
TinyBERT	$\mathbf{A}_f$	1	0.9571	0.9427	0.9672	18.1547
TinyBERT	$\mathbf{A}_f$	3	0.9549	0.9417	0.9602	16.4405
TinyBERT	$\mathbf{A}_q$	1	0.9534	0.9457	0.9525	14.8124
TinyBERT	attn_qkv	1	0.9513	0.9330	0.9500	13.4876
TinyBERT	$\mathbf{A}_q$	2	0.9410	0.9210	0.9560	15.7367
TinyBERT	$\mathbf{A}_k$	1	0.9334	0.9126	0.9467	14.3034
TinyBERT	$\mathbf{A}_k$	2	0.9159	0.9042	0.9192	16.0499
TinyBERT	$\mathbf{A}_v$	1	0.9155	0.9130	0.9185	14.4190
TinyBERT	$\mathbf{A}_v$	3	0.9067	0.7671	0.9162	12.3461
TinyBERT	attn_qkv	2	0.9043	0.9014	0.9068	11.2077
TinyBERT	$\mathbf{A}_f$	2	0.9015	0.8981	0.9236	19.2410
TinyBERT	$\mathbf{A}_p$	3	0.8850	0.8077	0.8977	14.6038
TinyBERT	attn_qkv	3	0.8795	0.7148	0.8860	12.5174
TinyBERT	$\mathbf{A}_q$	3	0.8585	0.6759	0.8776	14.0447
TinyBERT	$\mathbf{A}_k$	3	0.8429	0.6667	0.8905	15.1585
TinyBERT	$\mathbf{A}_p$	1	0.8296	0.7363	0.8618	13.0770
TinyBERT	$\mathbf{A}_p$	2	0.8180	0.7133	0.8460	13.4088
TinyBERT	$\mathbf{A}_p$	0	0.7557	0.6348	0.8099	15.9206
TinyBERT	$\mathbf{A}_o$	2	0.6548	0.5437	0.7976	13.1627
TinyBERT	$\mathbf{A}_o$	1	0.5823	0.4836	0.6888	15.1126
TinyBERT	$\mathbf{A}_o$	0	0.5455	0.3920	0.6681	17.0855
TinyBERT	$\mathbf{A}_o$	3	0.4852	0.2494	0.6125	15.4908
BERT_{FT}	$\mathbf{A}_k$	1	**0.9912**	0.9896	0.9910	10.3767
BERT_{FT}	$\mathbf{A}_q$	1	0.9815	0.9791	0.9805	11.9883
BERT_{FT}	$\mathbf{A}_v$	1	0.9277	0.8611	0.9412	10.9072
BERT_{FT}	$\mathbf{A}_f$	1	0.8757	0.7777	0.8979	18.0769

Table 3 – continued from previous page

Model	Target Component	Layer	ROUGE-L	METEOR	Token Acc	Loss
BERT_{FT}	$\mathbf{A}_p$	1	0.8695	0.8364	0.9060	16.4328
BERT_{FT}	`all_layers`	1	0.8423	0.7908	0.8865	11.4717
BERT_{FT}	`all_layers`	all	0.8344	0.7212	0.8617	11.4804
BERT_{FT}	$\mathbf{A}_o$	1	0.7792	0.7233	0.8320	14.8287
BERT_{FT}	$\mathbf{A}_q$	3	0.9119	0.8414	0.9233	12.3593
BERT_{FT}	$\mathbf{A}_k$	3	0.9087	0.8394	0.9267	13.0220
BERT_{FT}	`all_layers`	3	0.9040	0.8490	0.9233	11.7788
BERT_{FT}	$\mathbf{A}_p$	3	0.8847	0.7837	0.9035	15.8829
BERT_{FT}	$\mathbf{A}_v$	3	0.8344	0.7494	0.8601	10.4464
BERT_{FT}	$\mathbf{A}_f$	3	0.8053	0.7255	0.8692	19.2318
BERT_{FT}	$\mathbf{A}_o$	3	0.6950	0.6060	0.7945	12.7738

Table 4: Reconstruction metrics on the **CoLA** dataset across TinyBERT and BERT_{FT}.

Model	Target Component	Layer	ROUGE-L	METEOR	Token Acc	Loss
TinyBERT	$\mathbf{A}_k$	0	**1.0000**	1.0000	1.0000	13.8437
TinyBERT	$\mathbf{A}_q$	0	**1.0000**	1.0000	1.0000	11.9469
TinyBERT	`attn_qkv`	0	**1.0000**	1.0000	1.0000	11.2659
TinyBERT	$\mathbf{A}_v$	1	**1.0000**	1.0000	1.0000	14.2597
TinyBERT	$\mathbf{A}_v$	0	**1.0000**	1.0000	1.0000	10.6858
TinyBERT	$\mathbf{A}_f$	3	**0.9956**	0.9825	0.9949	15.7245
TinyBERT	$\mathbf{A}_f$	2	**0.9956**	0.9709	0.9949	18.4014
TinyBERT	$\mathbf{A}_k$	1	**0.9953**	0.9823	0.9946	14.1781
TinyBERT	$\mathbf{A}_f$	1	**0.9952**	0.9906	0.9944	17.2864
TinyBERT	$\mathbf{A}_v$	2	**0.9934**	0.9725	0.9950	13.1020
TinyBERT	`attn_qkv`	2	**0.9919**	0.9747	0.9928	11.1056
TinyBERT	$\mathbf{A}_q$	2	0.9895	0.9467	0.9907	15.1732
TinyBERT	$\mathbf{A}_q$	1	0.9887	0.9764	0.9869	14.9335
TinyBERT	`attn_qkv`	1	0.9871	0.9697	0.9851	13.0758
TinyBERT	$\mathbf{A}_f$	0	0.9867	0.9674	0.9847	15.7064
TinyBERT	$\mathbf{A}_k$	2	0.9755	0.9446	0.9763	15.8220
TinyBERT	$\mathbf{A}_k$	3	0.9703	0.9083	0.9690	15.1760
TinyBERT	`attn_qkv`	3	0.9536	0.8802	0.9742	12.5355
TinyBERT	$\mathbf{A}_v$	3	0.9360	0.8287	0.9740	11.5253
TinyBERT	$\mathbf{A}_q$	3	0.9102	0.7586	0.9609	13.8783
TinyBERT	$\mathbf{A}_p$	0	0.8958	0.8399	0.8973	15.9019
TinyBERT	$\mathbf{A}_p$	3	0.8928	0.8289	0.9165	14.3622
TinyBERT	$\mathbf{A}_p$	2	0.8835	0.7885	0.9201	13.7221
TinyBERT	$\mathbf{A}_p$	1	0.8463	0.6462	0.8936	13.2495
TinyBERT	$\mathbf{A}_o$	2	0.8124	0.6401	0.8828	14.1759
TinyBERT	$\mathbf{A}_o$	1	0.7417	0.6522	0.8499	14.7338
TinyBERT	$\mathbf{A}_o$	0	0.7077	0.5187	0.8013	16.4065
TinyBERT	$\mathbf{A}_o$	3	0.5790	0.3482	0.7505	16.4107
BERT_{FT}	`all_layers`	1	**1.0000**	1.0000	1.0000	11.7394
BERT_{FT}	$\mathbf{A}_p$	1	**1.0000**	1.0000	1.0000	17.3997
BERT_{FT}	$\mathbf{A}_k$	1	**1.0000**	1.0000	1.0000	10.4067
BERT_{FT}	$\mathbf{A}_q$	1	**1.0000**	1.0000	1.0000	10.8682
BERT_{FT}	`all_layers`	all	**0.9970**	0.9949	0.9938	11.1376
BERT_{FT}	$\mathbf{A}_f$	1	0.9668	0.9001	0.9673	17.7219
BERT_{FT}	$\mathbf{A}_v$	1	0.9635	0.9268	0.9529	10.3556
BERT_{FT}	$\mathbf{A}_o$	1	0.8838	0.8070	0.9095	15.4895
BERT_{FT}	$\mathbf{A}_k$	3	**1.0000**	1.0000	1.0000	13.4392
BERT_{FT}	$\mathbf{A}_q$	3	**1.0000**	1.0000	1.0000	12.2279
BERT_{FT}	$\mathbf{A}_v$	3	**1.0000**	1.0000	1.0000	10.7323
BERT_{FT}	`all_layers`	3	**0.9923**	0.9919	0.9968	13.1418
BERT_{FT}	$\mathbf{A}_f$	3	0.9852	0.9587	0.9904	18.8925
BERT_{FT}	$\mathbf{A}_p$	3	0.9752	0.9643	0.9857	15.6063
BERT_{FT}	$\mathbf{A}_o$	3	0.9271	0.8544	0.9424	12.3401

Table 5: Reconstruction metrics on the **Rotten Tomatoes** dataset across TinyBERT and BERT_{FT}.

Model	Target Component	Layer	ROUGE-L	METEOR	Token Acc	Loss
TinyBERT	$\mathbf{A}_q$	0	**0.9984**	0.9935	0.9981	7.1141
TinyBERT	$\mathbf{A}_k$	0	**0.9968**	0.9897	0.9963	8.1762
TinyBERT	$\mathbf{A}_q$	1	**0.9965**	0.9423	0.9959	8.8361

Table 5 – continued from previous page

Model	Target Component	Layer	ROUGE-L	METEOR	Token Acc	Loss
TinyBERT	attn_qkv	1	**0.9950**	0.9559	0.9963	7.7985
TinyBERT	$\mathbf{A}_k$	1	**0.9950**	0.9668	0.9943	8.6508
TinyBERT	attn_qkv	0	**0.9950**	0.9836	0.9963	6.7908
TinyBERT	$\mathbf{A}_v$	1	**0.9937**	0.9682	0.9930	8.0769
TinyBERT	$\mathbf{A}_q$	2	**0.9920**	0.9190	0.9908	9.6641
TinyBERT	$\mathbf{A}_f$	3	**0.9919**	0.9439	0.9926	9.0900
TinyBERT	$\mathbf{A}_v$	2	**0.9909**	0.9769	0.9896	7.7858
TinyBERT	attn_qkv	2	**0.9905**	0.9609	0.9892	6.7266
TinyBERT	$\mathbf{A}_v$	0	**0.9903**	0.9744	0.9910	6.5218
TinyBERT	$\mathbf{A}_f$	1	0.9875	0.9637	0.9859	10.2993
TinyBERT	$\mathbf{A}_f$	2	0.9874	0.9002	0.9893	10.9620
TinyBERT	$\mathbf{A}_p$	0	0.9849	0.9267	0.9765	9.9501
TinyBERT	$\mathbf{A}_k$	2	0.9823	0.8737	0.9815	9.3244
TinyBERT	$\mathbf{A}_f$	0	0.9803	0.9136	0.9814	9.5539
TinyBERT	$\mathbf{A}_k$	3	0.9758	0.8441	0.9874	9.1323
TinyBERT	$\mathbf{A}_p$	3	0.9715	0.8361	0.9751	9.2380
TinyBERT	$\mathbf{A}_q$	3	0.9689	0.8458	0.9870	8.3287
TinyBERT	$\mathbf{A}_v$	3	0.9634	0.9308	0.9963	7.6839
TinyBERT	$\mathbf{A}_p$	2	0.9589	0.7661	0.9679	8.3896
TinyBERT	$\mathbf{A}_p$	1	0.9584	0.6959	0.9562	8.2822
TinyBERT	attn_qkv	3	0.9380	0.8525	0.9846	7.7289
TinyBERT	$\mathbf{A}_o$	1	0.9154	0.6596	0.9469	9.8757
TinyBERT	$\mathbf{A}_o$	0	0.8603	0.5743	0.9328	10.2975
TinyBERT	$\mathbf{A}_o$	2	0.8483	0.5547	0.9363	10.1219
TinyBERT	$\mathbf{A}_o$	3	0.7366	0.4225	0.9061	10.6028
BERT_{FT}	$\mathbf{A}_o$	1	**0.9984**	0.9341	0.9981	9.6163
BERT_{FT}	$\mathbf{A}_k$	1	**0.9984**	0.9894	0.9981	6.3293
BERT_{FT}	$\mathbf{A}_q$	1	**0.9968**	0.9912	0.9963	6.9075
BERT_{FT}	all_layers	1	**0.9965**	0.9835	0.9961	6.7434
BERT_{FT}	$\mathbf{A}_p$	1	**0.9900**	0.9607	0.9904	7.8044
BERT_{FT}	all_layers	all	0.9889	0.9796	0.9856	6.5090
BERT_{FT}	$\mathbf{A}_v$	1	0.9838	0.9467	0.9795	6.2343
BERT_{FT}	$\mathbf{A}_f$	1	0.9703	0.8164	0.9721	11.0523
BERT_{FT}	$\mathbf{A}_k$	3	**0.9965**	0.9841	0.9961	8.3351
BERT_{FT}	$\mathbf{A}_v$	3	**0.9965**	0.9499	0.9923	5.9356
BERT_{FT}	$\mathbf{A}_q$	3	**0.9950**	0.9826	0.9943	6.8756
BERT_{FT}	all_layers	3	**0.9927**	0.9746	0.9961	7.3515
BERT_{FT}	$\mathbf{A}_f$	3	0.9768	0.8189	0.9755	11.7488
BERT_{FT}	$\mathbf{A}_o$	3	0.9722	0.8546	0.9861	7.8487
BERT_{FT}	$\mathbf{A}_p$	3	0.9719	0.9284	0.9923	8.9211

Table 6: Activation sizes for $\text{BERT}_{\text{BASE}}$ and TinyBERT. Counts are reported per token position; multiply by $B \times S$ for the total number of scalar activations in a batch. The used ratio is computed with respect to all target activations, including the embedding output and all target transformer sub-module activations.

Model	Employed Activations	Notation	# Activations	Used Ratio %
$\text{BERT}_{\text{BASE}}$ ($L = 12, H = 768, I = 3072$)				
$\text{BERT}_{\text{BASE}}$	All Target Activations (Baseline)	A_{all}	83,712	100.00
$\text{BERT}_{\text{BASE}}$	All Transformer Activations	A_T	82,944	99.08
$\text{BERT}_{\text{BASE}}$	i-th Transformer Layer Activations	$A_{t,i}$	6,912	8.26
$\text{BERT}_{\text{BASE}}$	i-th FFN Fully Connected	$A_{f,i}$	3,072	3.67
$\text{BERT}_{\text{BASE}}$	i-th FFN Output	$A_{p,i}$	768	0.92
$\text{BERT}_{\text{BASE}}$	i-th Attention Output	$A_{o,i}$	768	0.92
$\text{BERT}_{\text{BASE}}$	i-th Attention Query	$A_{q,i}$	768	0.92
$\text{BERT}_{\text{BASE}}$	i-th Attention Key	$A_{k,i}$	768	0.92
$\text{BERT}_{\text{BASE}}$	i-th Attention Value	$A_{v,i}$	768	0.92
TinyBERT ($L = 4, H = 312, I = 1200$)				
TinyBERT	All Target Activations (Baseline)	A_{all}	11,352	100.00
TinyBERT	All Transformer Activations	A_T	11,040	97.25
TinyBERT	i-th Transformer Layer Activations	$A_{t,i}$	2,760	24.31
TinyBERT	i-th FFN Fully Connected	$A_{f,i}$	1,200	10.57
TinyBERT	i-th FFN Output	$A_{p,i}$	312	2.75
TinyBERT	i-th Attention Output	$A_{o,i}$	312	2.75
TinyBERT	i-th Attention Query	$A_{q,i}$	312	2.75
TinyBERT	i-th Attention Key	$A_{k,i}$	312	2.75
TinyBERT	i-th Attention Value	$A_{v,i}$	312	2.75